Data Collection and Preprocessing Phase

Field	Detail
Student Name	Dinesh
Date	19 July 2026
Team ID	Individual
Project Title	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Maximum Marks	3 Marks

Data Quality Report Template

This Data Quality Report summarizes the quality assessment of the global AI-in-education dataset used in the project *Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report*. It documents identified data quality issues, their severity levels, and the resolution plans applied to ensure accurate, consistent, and reliable data for Tableau analysis and dashboard development.

Data Source	Data Quality Issue	Severity	Resolution Plan
Global_AI_in_Education.csv	15 of 16 columns fully populated with zero missing values (year, month, country, region, student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct, avg_daily_ai_usage_min, top_ai_tool, internet_penetration_pct, education_index, urban_ai_usage_pct, rural_ai_usage_pct, ai_in_curriculum, gender_gap_ai_usage_pct).	Low	No action required. Verified using a nullcount check across all columns — result: 0 nulls in 15 of 16 fields.
Global_AI_in_Education.csv	government_ai_policy contains 441 missing values (32.4% of 1,360 records). These represent country/period	Moderate	Replace nulls with an explicit “None” category using IFNULL()/COALESCE so all three policy states — None, Draft, Active — are

	combinations with no formal AI-in-education		

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	policy rather than a datacollection error.		represented consistently in filters, tooltips, and color legends.
Global_AI_in_Education.csv	No duplicate records found. All 1,360 countryyear-month combinations are unique with zero duplication across the entire dataset.	Low	No action required. Duplicate check on the country-yearmonth key returned 0 duplicates. Dataset integrity confirmed for all records.
Global_AI_in_Education.csv	Region-level totals are unevenly weighted: Asia contains 408 records (3 countries — China, India, Pakistan) while Africa, South America, and Oceania each contain 136 records (1 country each). A simple AVG() by region can misrepresent regional performance.	Moderate	Add a per-country average measure alongside the region total, and label regionlevel charts clearly as “country-weighted” where relevant. Consider a parameter to toggle between SUM and per-country AVG views.
Global_AI_in_Education.csv	education_index is stored as float64 but ranges narrowly between 0.52 and 0.94, which can visually flatten differences between countries on a default 0–1 axis.	Low	Set a custom axis range (e.g. 0.4–1.0) or add reference lines/bands in Tableau to preserve visible differentiation between countries.

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Global_AI_in_Education.csv	year and month are stored as separate int64 columns. A month value alone (e.g. “6”) is ambiguous across the 2015–2026 span and may mislead trend charts if plotted independently of year.	Low	Create a composite calculated field Period = STR([year]) + “-” + RIGHT(“0” + STR([month]), 2), formatted as YYYYMM, for use on all time-series axes.
Global_AI_in_Education.csv	ai_in_curriculum is stored as “Yes”/“No” text rather than a boolean flag, which reduces readability for rate calculations and nontechnical dashboard viewers.	Low	Create a Tableau calculated field: IF [ai_in_curriculum] = “Yes” THEN 1 ELSE 0 END, and use it to compute a Curriculum Integration Rate for KPI cards and trend lines.
Global_AI_in_Education.csv	top_ai_tool records only a single leading tool per country-month (ChatGPT, Google Gemini, Microsoft Copilot, or Khanmigo), which limits analysis of tool co-adoption or secondary-tool usage.	Low	Document this as a stated assumption in the project README: the field represents the leading tool only, not the full tool stack in use. No further correction possible without additional source data.

Global_AI_in_Education.csv	No standardized composite keys exist for cross-dimensional analysis (e.g. no single column combining country + region, or government_ai_policy + ai_in_curriculum), limiting direct multiattribute segmentation.	Low	Create derived calculated columns in Tableau: Location Label = [country] + ” - ” + [region]; Policy-Curriculum Alignment = [government_ai_policy (clean)] + ” - ” + [ai_in_curriculum]. Use these for heatmaps and crosssegment filters.
Global_AI_in_Education.csv	urban_ai_usage_pct and rural_ai_usage_pct are independent columns (ranges 0–70 and 0–59 respectively); the derived Urban-Rural Usage Gap could theoretically be negative, which would need review rather than being treated as an error by default.	Low	Add a calculated field Urban-Rural Usage Gap = [urban_ai_usage_pct] – [rural_ai_usage_pct], validate the sign distribution, and flag any negative values for manual review rather than automatic exclusion.